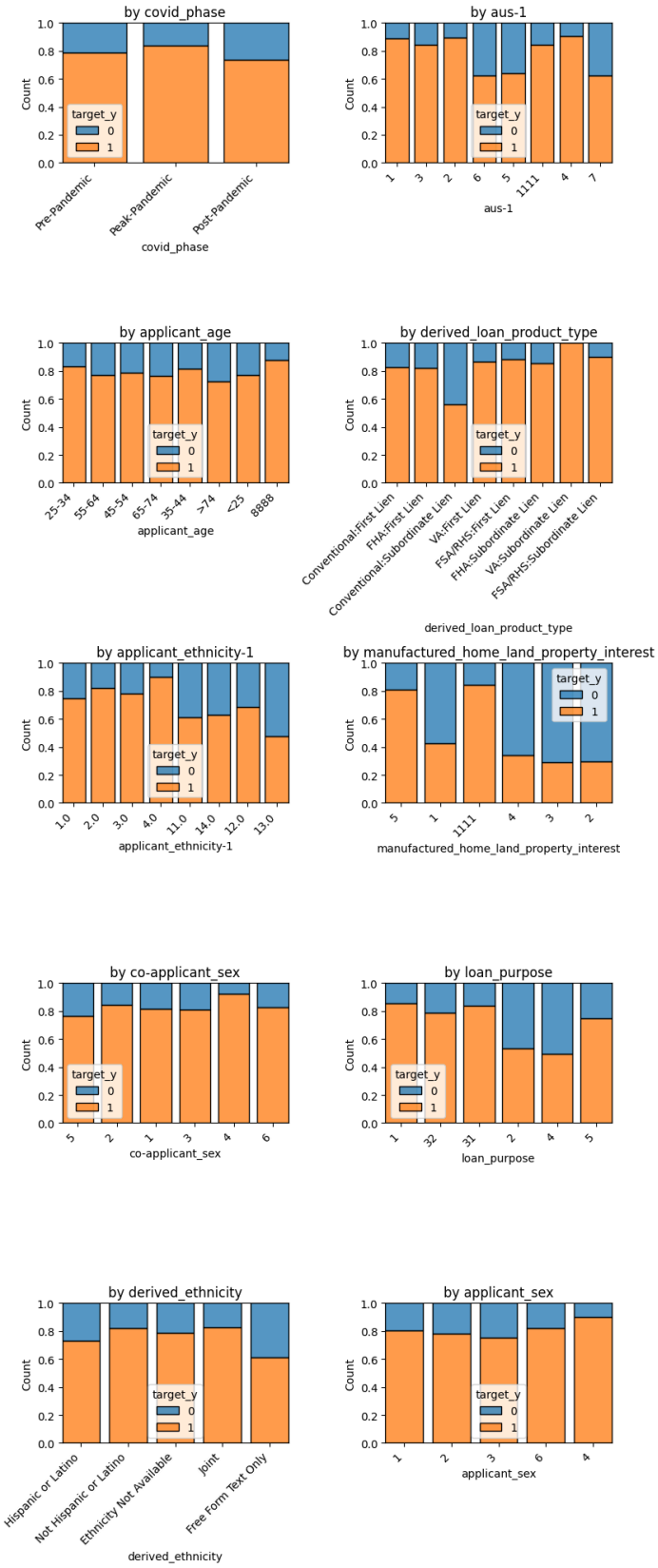
Appendix

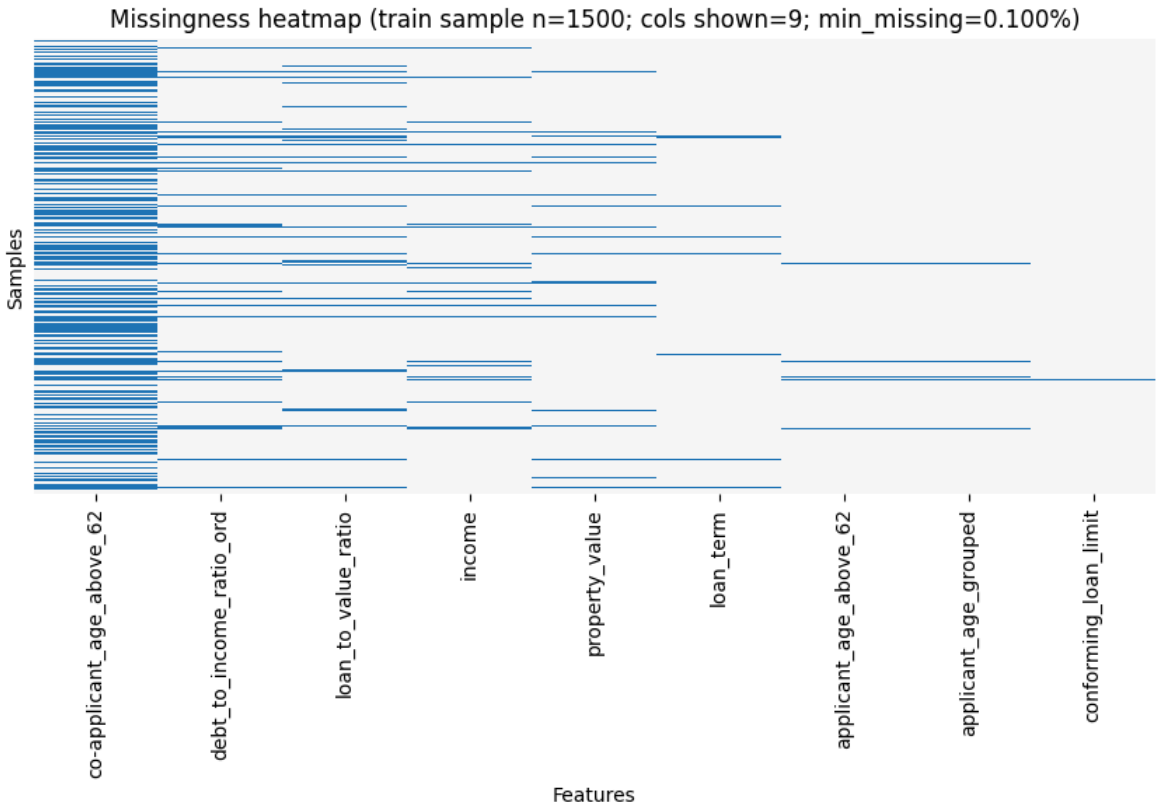
1. Feature Consolidation Details (2.2.3)

Detailed distribution charts



Based on the visual EDA of approval rates across the isolated categorical features, we executed the following consolidation strategy to maximize predictive signal while minimizing dimensionality:

- **Binarized by Presence/Type:** * `co-applicant_sex` : Gender showed zero variance in approval outcomes. The true signal was simply the presence of a co-applicant. Converted to a binary `is_joint_application` (0 = Single, 1 = Joint).
  - `derived_loan_product_type` : Grouped into `First_Lien` (>80% approval) vs. `Subordinate_Lien` (~50% approval).
- **Binned by Risk/Approval Rates:**
  - `aus-1` : Grouped official systems into `Standard_AUS` (codes 1-4, ~90% approval) and mapped the rest to `Non_Standard_or_Exempt` (~60% approval).
  - `loan_purpose` : Grouped standard applications (purchase, refi) into `Standard_Purpose` , and higher-denial applications (home improvement, other) into `High_Risk_Purpose` .
  - `manufactured_home_land_property_interest` : Consolidated into `Has_Property_Interest` and `Not_Applicable_or_Exempt` based on distinct denial rate clusters.
- **Demographic Bucketing:** `applicant_age` was collapsed into broader life-stages (`<25` , `25-44` , `>44`). Anomaly placeholder values (8888, 9999) were explicitly replaced with `NaN` for safe downstream imputation.
- **Dropped:** `applicant_sex` (perfectly flat approval rates across categories; zero predictive value) and `applicant_ethnicity-1` (100% redundant with `derived_ethnicity`).

2. Imputation (2.3)

Heatmap before Imputation